Ethan Weimer-Kopf

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About

Mechanical engineering student drawn to the problems at the intersection of thermal, fluids, and materials engineering. I currently work on various teams of engineers on Northwestern's Formula SAE and Space Technology teams to design, test, and build systems that perform under strict physical limitations and constraints.

Skills and Tools

CAD / Design: SolidWorks, Fusion360, Onshape; GD&T (ASME Y14.5 – 2018)

Analysis: SolidWorks FEA, Mass, and Motion Analysis; Onshape thermal and structural analysis

Fabrication: 3D printing; Manual and CNC Mill and Lathe

Programming: MATLAB, Python, Java

Northwestern Formula Racing (NFR) 2027 Hubs Design

Club Project | Summer 2026 - Present | Tools: SolidWorks (CAD, FEA, CAM)

Problem: *The NFR 2027 Suspension system is undergoing a complete overhaul to 4 in hub motors, and with it a completely different corner architecture. The hubs, which transfer torque from the motor to the wheel, must now be completely redesigned to a much smaller size that connects to both the brake rotor and wheel, but also packages a ring gear (which transfers torque), two angular bearings, and a lip seal and gasket.*

My Role: *Gathered constraints and dimensions, create design iterations accounting for manufacturability and cost, perform FEA testing to ensure proper strength and conduct mass optimization (Minimizing unsprung mass). Ensure seamless integration with the gear carriers, uprights, hub motors, brake rotors, and wheels.*

Process:

- *Used suspension geometry to create initial design space; create shoulders and appropriate fillets for bearings and their locations.*
- *Once initial geometry was created, different mass cuts were made, and FEA (using known max cornering, breaking, and acceleration loads) was used to determine the optimal cuts that maintain performance while cutting the most mass.*
- *Integrated into assembly with other parts, using GD&T to thoroughly ensure part compatibility when manufacturing.*

Outcome: Still in production; so far, I have managed to cut the model's mass down by ~45% from the initial geometry while maintaining structural integrity.

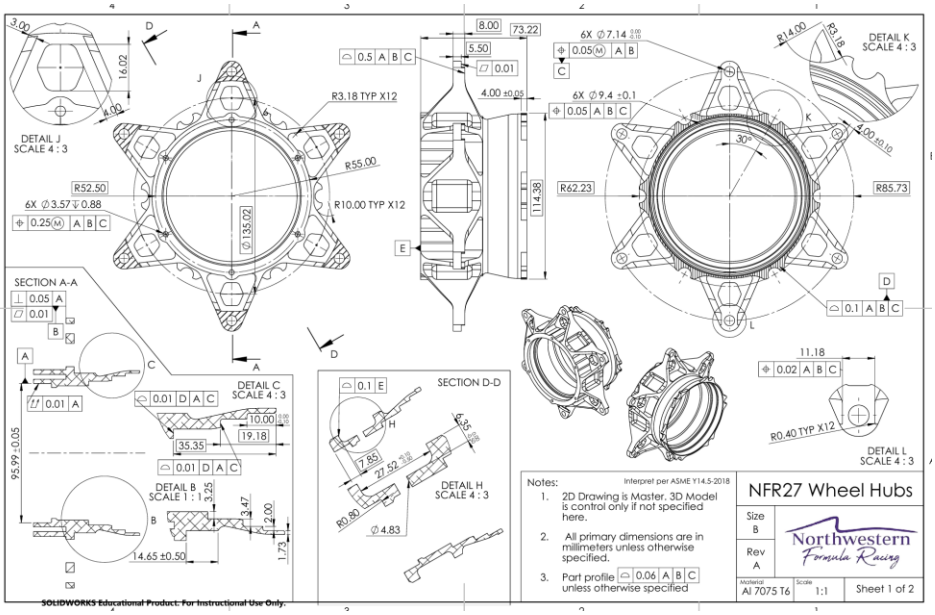


Figure 1: Hubs GD&T Design Drawing

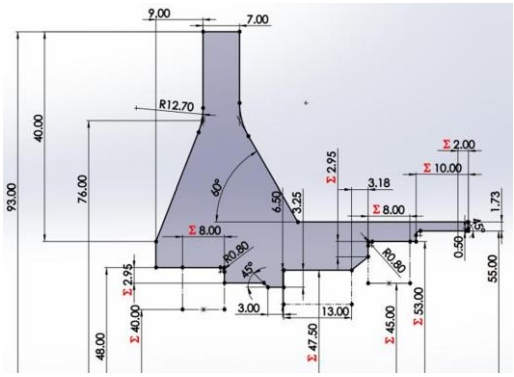


Figure 2: Design Space and Essential Geometry

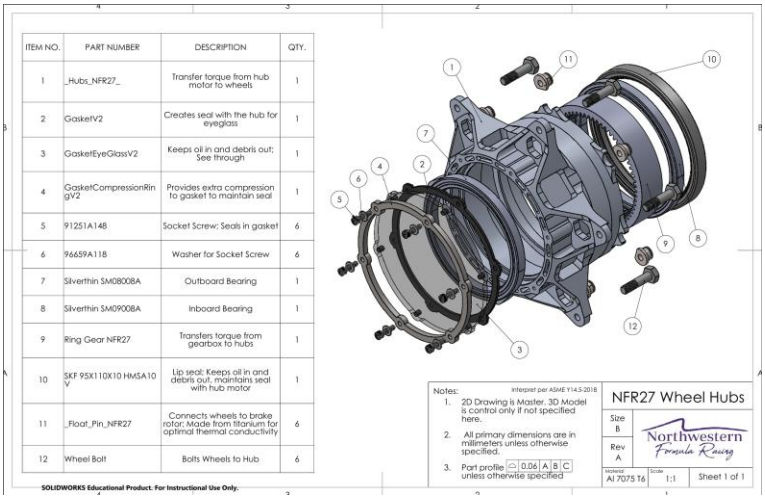


Figure 3: Full Hub Exploded View & BOM

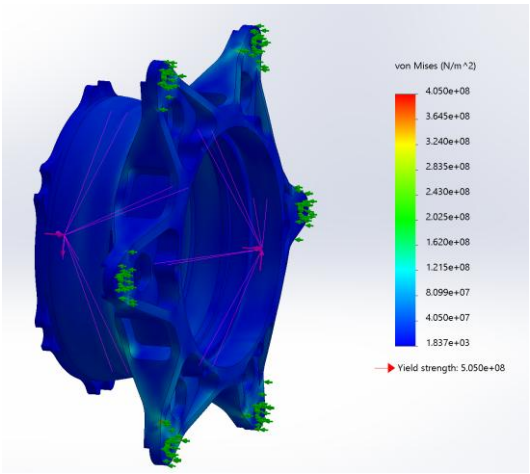


Figure 4: Hubs Outside Cornering FEA Case (Stress)

Liquid Rocket Engine Nozzle

Club Project | Spring 2026 - Present | Tools: Onshape (CAD), Ansys (Thermal Analysis), SolidWorks (Drawings), CNC Lathe

Problem: Research, design, and manufacturing of the nozzle for Northwestern's first liquid rocket engine. The nozzle must withstand extreme temperatures and pressures, and its inner contour must follow precisely calculated geometry to produce parallel Mach lines and uniform exit flow, maximizing thrust.

My Role: I developed manufacturing tolerances and GD&T for the nozzle's graphite throat insert, which sees the highest temperatures and pressures and must be toleranced precisely to account for thermal expansion. I am also machining the aluminum outer shell on a CNC lathe, and I will support hydrostatic testing and test fires once the parts are complete.

Process:

- Materials selection based on minimizing thermal shock and creep. Graphite throat insert chosen due to its high heat resistance and consistent yield strength at high temperatures Aluminum outer shell chosen to maximize heat dissipation.
- Tolerances for throat insert decided by analyzing differences in maximum thermal expansion between the throat insert and outer aluminum shell. Medium interference fit chosen to maintain firm fit at maximum thermal expansion while keeping installation force low.

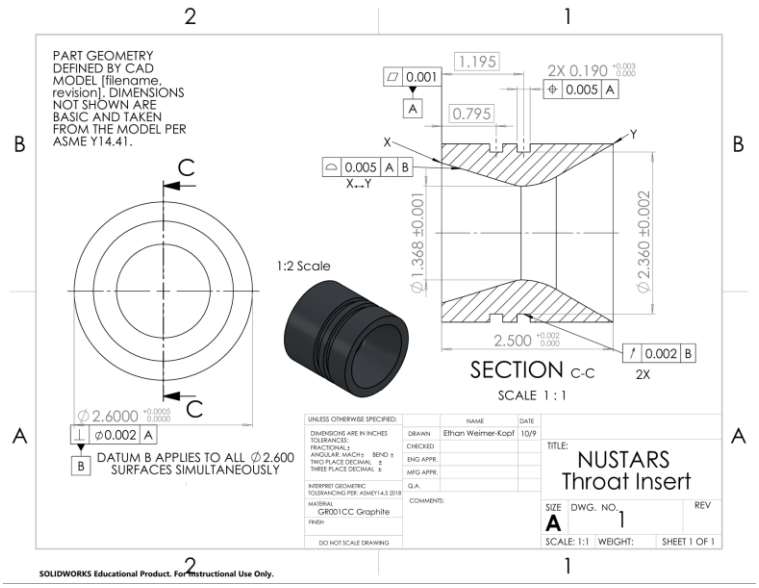


Figure 5: Graphite Throat Insert GD&T

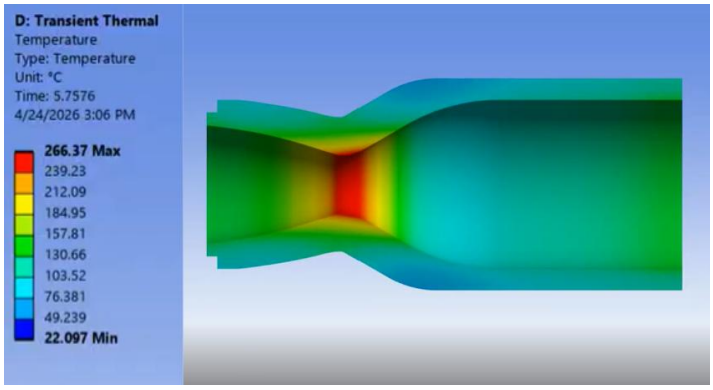


Figure 6: Onshape thermal testing

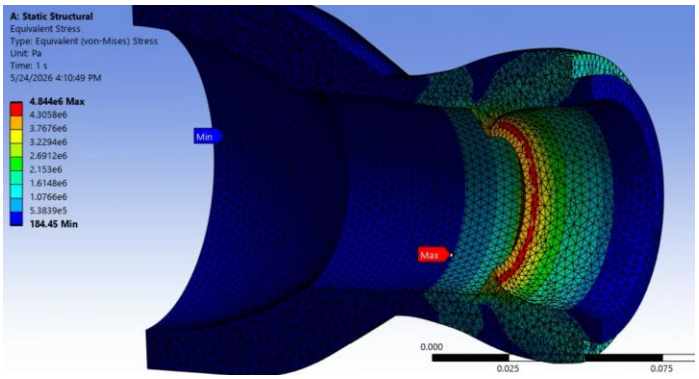


Figure 7: Onshape structural testing

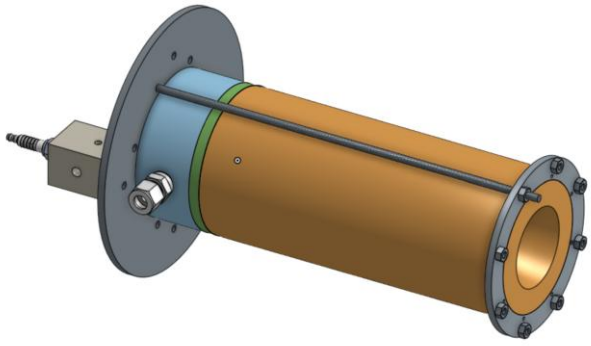


Figure 8: Nozzle, Combustion Chamber, and Igniter Assembly

Rear Axle Dampening System: Intra-Tripod Spring Plunger (TSP)

Club Project | Winter-Spring 2026 | Tools: SolidWorks (CAD and FEA), Manual and CNC Lathe

Problem: *The rear half shafts, which transfer torque from the drivetrain to the hubs for NFR26, exhibited excessive axial play within the tripod housings, allowing uncontrolled movement and impact damage during cornering loads. I was tasked with designing a damping system to dissipate this kinetic energy and maintain constant axial contact between the half shaft and tripod housing.*

My Role: *Modeled assembly in SolidWorks, calculated loads and selected material. Physically constructed components (plug and plunger) on the CNC lathe.*

Process:

- *Analyzed centrifugal loads at cornering to determine optimal spring constant, spring length, and compression at maximum load (2x factor of safety used) so that maximum compression occurs at maximum load.*
- $30\frac{N}{m}$ *spring and CV grease create a damped oscillating system that dissipates energy; Plunger at the end of the spring maintains constant axial contact with tripod housing.*

Outcome: *System was installed easily and functioned as intended. Rear Axles demonstrated sufficient and controlled axial play*

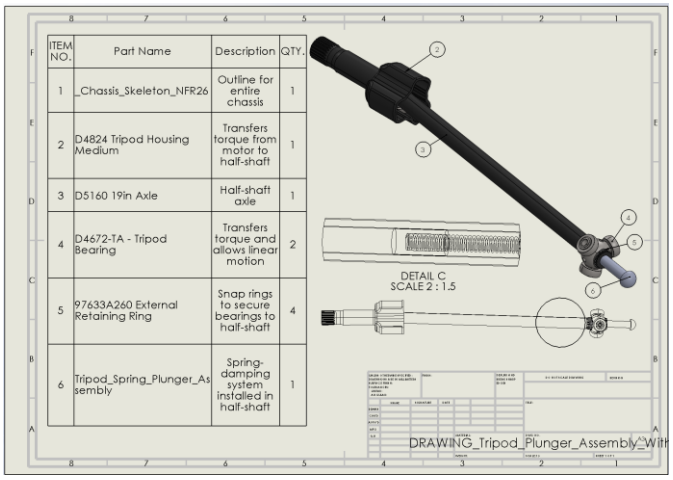


Figure 9: Half Shaft Drawing and BOM, including TSP.

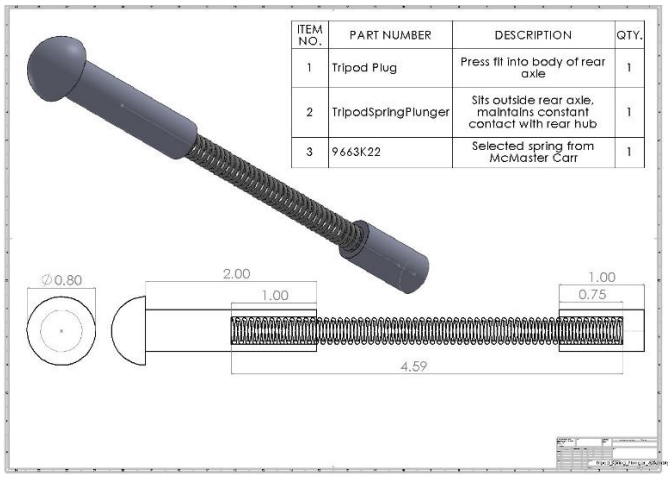


Figure 10: TSP Dimensioned Assembly



Figure 11: TSP Construction

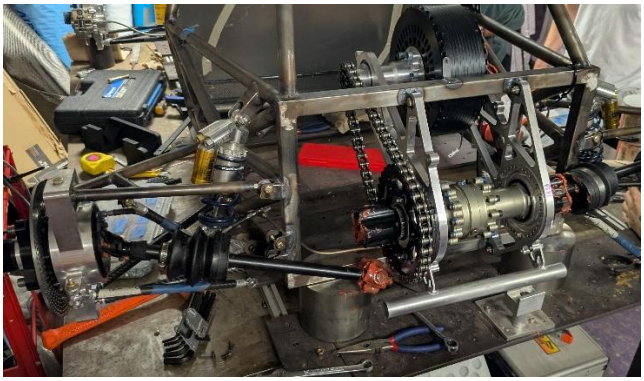


Figure 12: Integrating TSP into rear drive train

Custom Low - Torque Putter Build

Personal Project | Spring 2026 | Tools: Solidworks, BambuLab 3D printing

Goal: Create a prototype modular zero torque putter that can easily adjust lie angle and shaft length

My Role: Solo project, so I brainstormed the mechanism and design, modeled and ran analysis in Solidworks, and constructed it using shop tools, creating multiple design iterations and prototypes before arriving on the final version.

Process:

- *Steel and aluminum weights used to optimize mass and weight distribution of the putter. Weights were densely packed behind aluminum face insert to maximize inertia behind the point of contact with the ball.*
- SolidWorks mass properties analysis used to strategically place the center of mass. Variety of shaft inserts allow user to change the angle that the shaft (AKA the lie angle). All inserts are carefully designed to align the shaft axis directly through the center of mass of the putter, allowing for zero net torque about the shaft axis and therefore a much more controlled putting stroke.
- Motion analysis ran to verify minimal torque was achieved. Compared to a sample of other putters to quantify improvement.

Outcome: *Successful first prototype that achieves desired goal of low torque about the shaft axis, with adjustable shaft angles.*

Future Improvements: Make the putter out of a solid block of aluminum using a CNC mill, increasing mass and improving weight distribution and overall feel



Figure 13: Putter down the line view



Figure 14: Putter overhead view

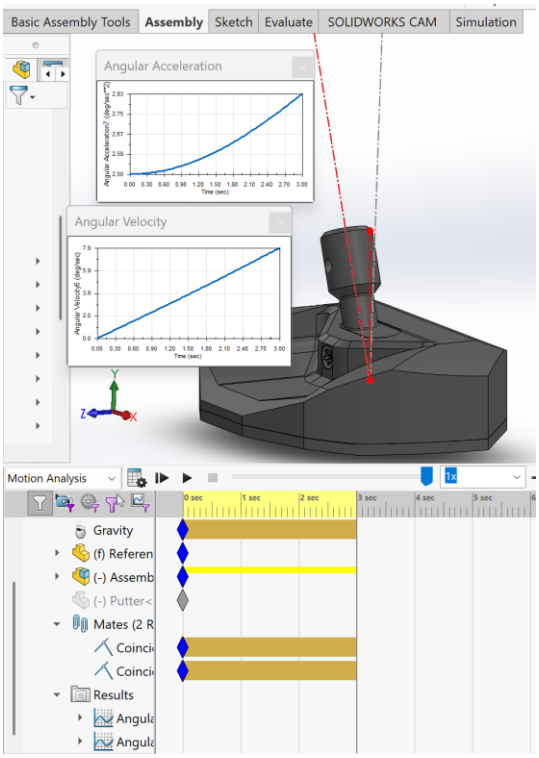


Figure 15: Motion Analysis Simulation

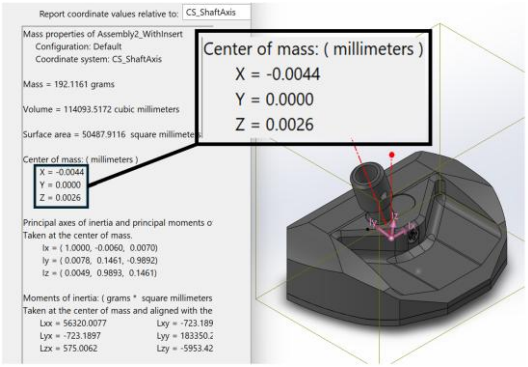
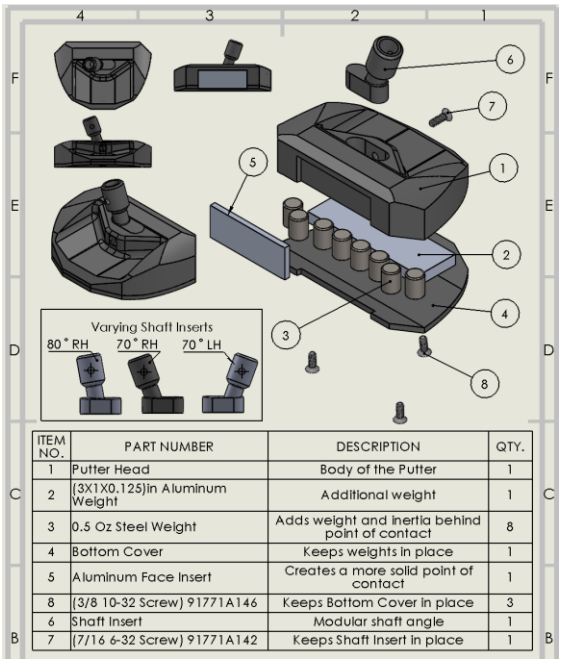


Figure 16: Mass Analysis on Final Product; Results show C.O.M very closely aligned with the shaft axis (~4 microns)



ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	Putter Head	Body of the Putter	1
2	(3X1X0.125)in Aluminum Weight	Additional weight	1
3	0.5 Oz Steel Weight	Adds weight and inertia behind point of contact	8
4	Bottom Cover	Keeps weights in place	1
5	Aluminum Face Insert	Creates a more solid point of contact	1
8	(3/8 10-32 Screw) 91771A146	Keeps Bottom Cover in place	3
6	Shaft Insert	Modular shaft angle	1
7	(7/16 6-32 Screw) 91771A142	Keeps Shaft Insert in place	1

Figure 17: Final Product, Optimal Weight Distribution and sufficient Mass (192g)

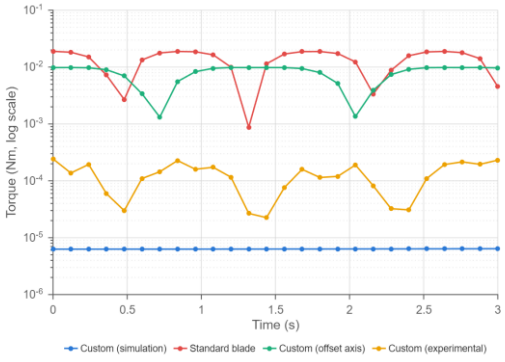


*Figure 18: **Simulated and measured torque vs. time over average stroke duration (3 s).** The custom putter showed markedly lower and more consistent torque than alternatives in simulation; experimental testing confirmed a significant reduction, though smaller than predicted theoretically.*

Wheelchair-Accessible Ball Launcher

Course Project | Spring 2026 | Tools: SolidWorks, BambuLab 3D printing, general machine shop tools

Problem: *Create a custom ball-launcher to help a 5th grader with spinal muscular atrophy play catch with his classmates*

My Role: *Sole CAD designer and 3D printer. Modeled and printed dozens of components custom fit to the client's wheelchair, iterated designs to optimize ease of interaction with the device, launch distance, durability, and customizability. Visited and interviewed the client in person to create problem constraints and gather feedback. Dimensioned all parts (see* Figure 20: Example of a Fully Dimensioned Part*) for co-authored 50-page related paper.*

Process:

- *~40 hours of CAD in Solidworks creating and testing dozens of parts across 3 main iterations before settling on final design.*
- *Plate and lever mechanism allows the client to activate the trigger despite his extremely limited range of motion*
- *Launches any ball from 2-8 inches in diameter; launches a standard whiffle ball 15-30 feet, depending on selected power level*
- *Custom wheelchair attachment allows an easy and secure setup as well as quick dismantling time*

Outcome: *Operated for 100+ launches without failure and made our client very happy!*

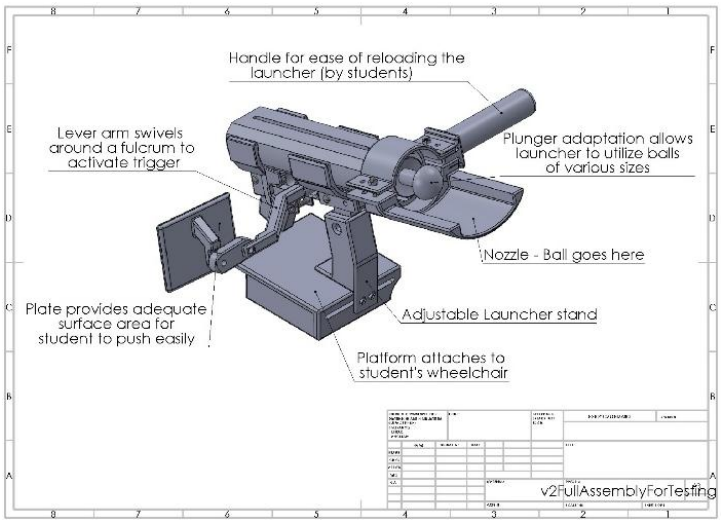


Figure 19: Isometric View and Descriptions

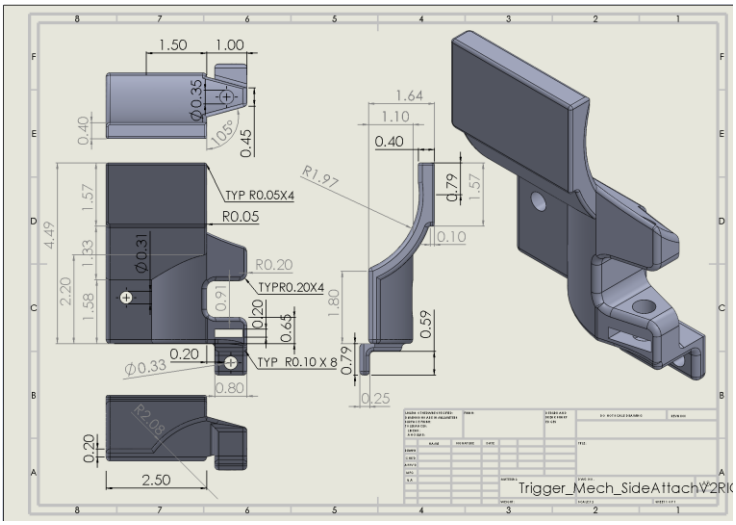


Figure 20: Example of a Fully Dimensioned Part



Figure 21: Final Product



Figure 22: Student using prototype during user testing

Portable and Modular Kiosk

Course Project | Winter 2026 | Tools: Onshape, Excel, Lathe, General shop tools (Band Saw, Panel Saw, Drill Press, etc)

Problem: *Small business vendors at pop-up markets often lack experience with in-person product display, and booth setup/teardown is slow and physically exhausting after a full day of selling. Our project partner, Claire Seizovic of ENSEMBLE (an Evanston indoor market for local businesses), asked us to design a modular, low-cost, transportable display booth that's easy to customize and quick to assemble/disassemble.*

My Role: *Interviewed 5 different vendors for required constraints and iterated models in Onshape. Built final prototype with teammate, and co-authored 90-page related research paper (detailing the full process, design, instructions for construction, interview summaries, and other relevant appendices)*

Process:

- *Fully collapsable structure, lightweight to promote ease of transport and ensure a <5-minute setup time.*
- *Created a back panel to lock kiosk and secure stored items; this panel when unfolded contains slats for lighting and display modules to increase customization and modularity.*
- *Wooden legs proved much too clumsy and heavy, so we allocated a large portion of our $150 budget to lightweight and sturdy aluminum legs and appropriate locking hinges to support them*

Outcome: *Held ~75lbs of weight easily for extended period and was fully assembled by 2 people in less than 20 seconds. Client chose our design as her favorite out of 4 total competing teams and took it with her for her business.*



Figure 23: Kiosk at Poster Fair Demonstration

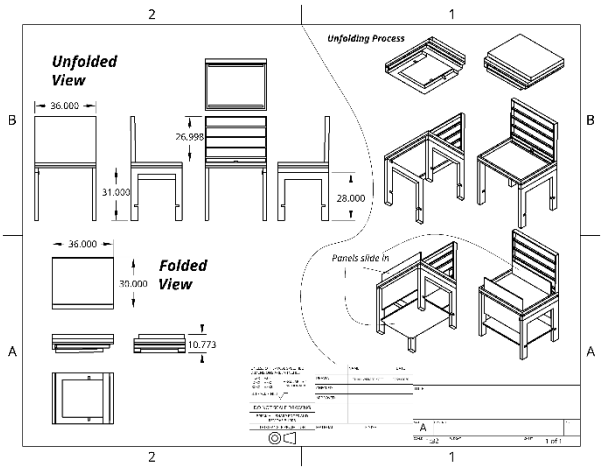


Figure 24: Kiosk Orthographic Drawing

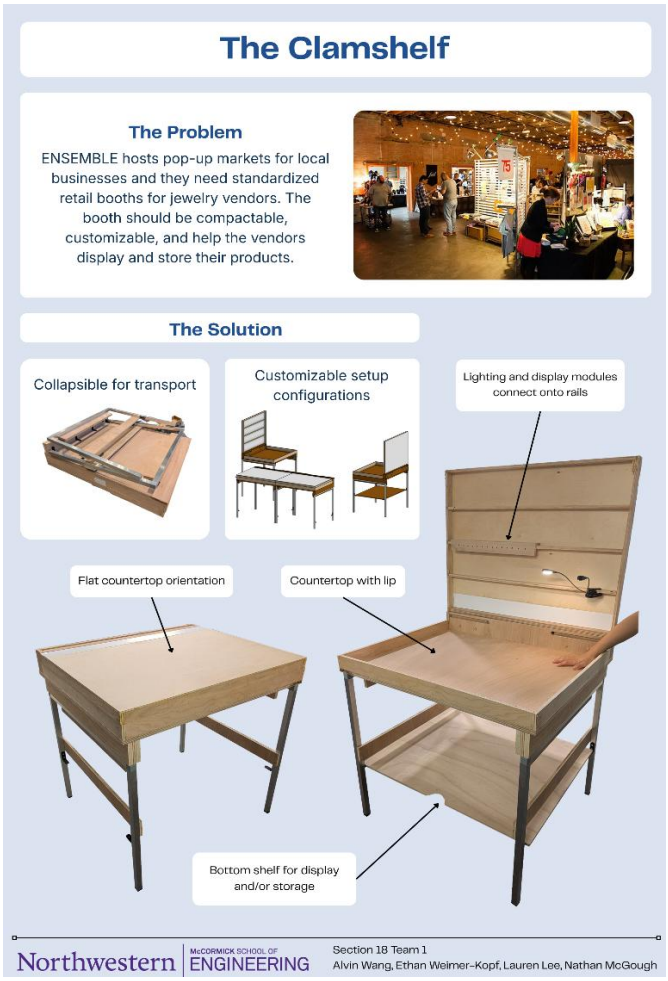


Figure 25: Kiosk Poster

Aerodynamic Modeling of Frisbee Golf Discs – International Physics Competition Champion

Course Project | Spring 2025 | Tools: Logger Pro, Excel, Vernier

Problem: *Investigate the aerodynamics of frisbee golf discs to create new characterization system based on real world data and experimentation.*

My Role: *Lead the research investigation, created testing setups, and collected and analyzed data. Interviewed professional frisbee golfers and talked with professors to narrow down the theoretical physics behind disc golf flight before testing theories experimentally with vernier force sensors and a custom "wind tunnel".*

Process:

- *Researched the physics of airfoils to better understand fluid forces and apply them to a frisbee model; interviewed professional disc golfers for insight on behavior of discs during flight and how different characteristics influence it.*
- *Used fans, vernier, logger pro, and available resources to set up extensive experiments to quantitatively measure the angle of attack force, Coanda effect, vortices, and Bernoulli's principle forces acting on dozens of different discs*
- *Compiled and analyzed data; discovered quantitative relationships of certain torques on the disc (for example, Fading Torque $\propto$ to $\sqrt{\theta\ (\text{Dome Angle})}$), and created custom characterization system based on the ratio between calculated torques causing turn and fade.*

Outcome: *Won the international tournament as a team; was offered an internship by one of the judges for the frisbee presentation.*

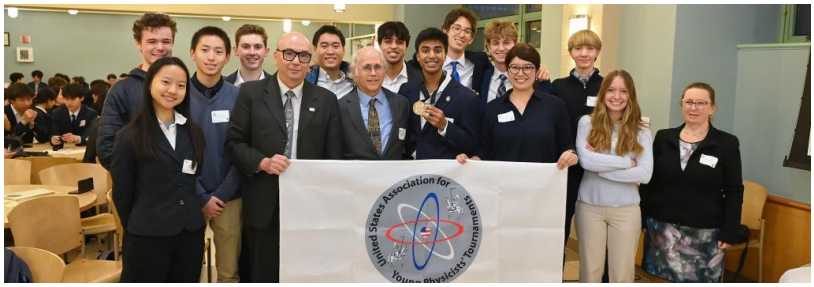


Figure 26: Team Photo!



Figure 27: Champion Certificate

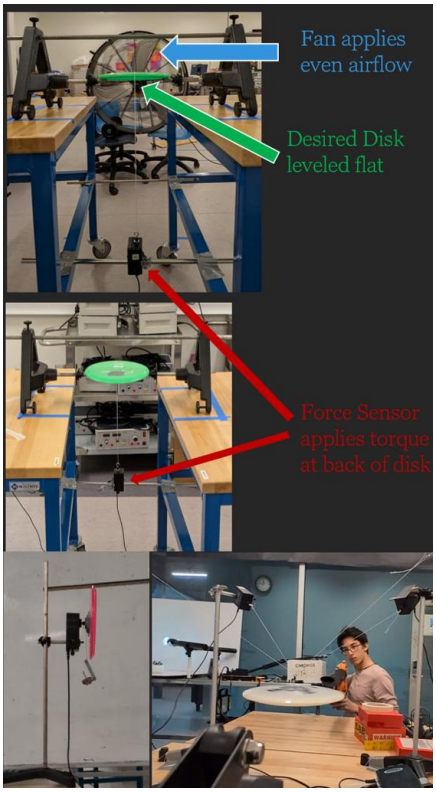


Figure 28: Experiment Setups

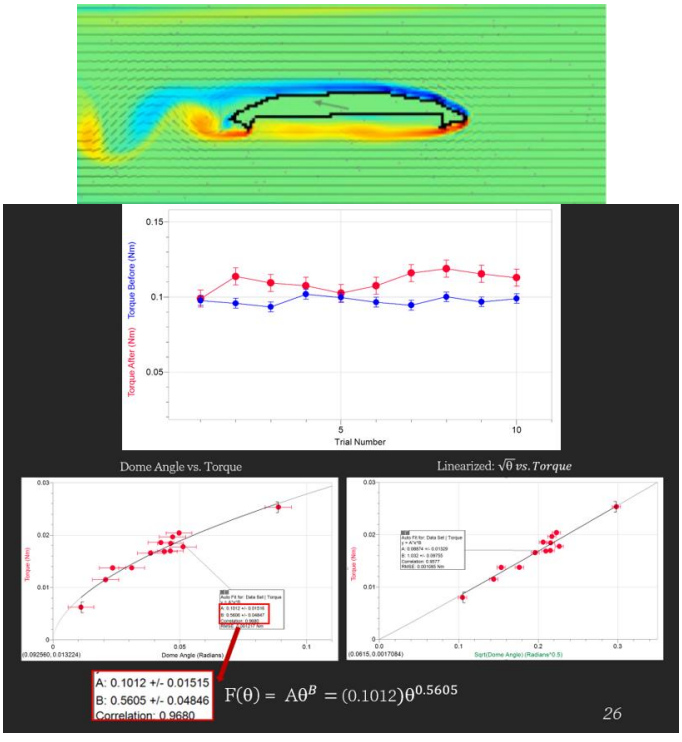


Figure 29: Some Experiment Results